



PSYCH 201B

Statistical Intuitions for Social Scientists

Power Review & Lab

*You can download these slides:
course website > Week 9 > Overview*

03/04/2025

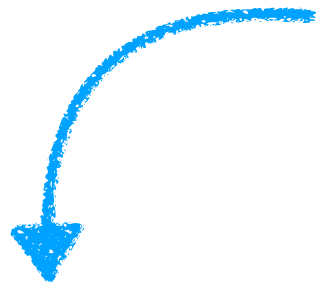
Announcements

- **March 12th (next Wed)**
 - Proposal Approval Deadline
 - You **must** meet with us **before** this to discuss proposal
- Today's plan
 - Mini-HW & Power Review
 - Readings and Lab Notebooks **Due Monday** (March 10th)

Mini-HW review

- Self-reported Accuracy (excluding Q13)
 - $M = 86.2\% \pm 10\%$
 - # Incorrect: 1.83 ± 1.21

This wording also confused me!



Q13: Is a statistically significant effect always a practically important effect?

- ☐ No, because in extremely large samples, extremely small effects can be statistically significant, and small effects are never practically important.
- ☐ No, because the alpha level could in theory be set to 0.20, and in that case a significant effect is not practically important.
- ☒ ☒ No, because how important an effect is depends on a cost-benefit analysis, not on how surprising the data is under the null hypothesis.
- ☐ All of the above are true.

Mini-HW review

- Self-reported Accuracy (excluding Q13)
 - $M = 86.2\% \pm 10\%$
 - # Incorrect: 1.83 +/- 1.21

Q8: Let's assume that the random number generator in R works, and we use `rnorm(n = 50, mean = 0, sd = 1)` to generate 50 observations, and the mean of these observations is 0.5, which in a one-sample *t*-test against an effect of 0 yields a *p*-value of 0.03, which is smaller than the alpha level (which we have set to 0.05). What is the probability that we have observed a significant difference ($p < \alpha$) just by chance?

- ☐ 3%
- ☐ 5%
- ☐ 95%
- ☒  100%

Power Review

Imagine you're a (medical) doctor...

H_0 : Not pregnant H_1 : Pregnant

false
positive

Type I error

$p(\text{reject } H_0 | H_0 \text{ is true})$

α

false
negative

Type II error

$p(\text{not reject } H_0 | H_1 \text{ is true})$

β

true
positive

Sensitivity

$p(\text{reject } H_0 | H_1 \text{ is true})$

Power

$1 - \beta$

true
negative

$p(\text{not reject } H_0 | H_0 \text{ is true})$

Specificity

$1 - \alpha$

Ways to
be wrong



Ways to
be right

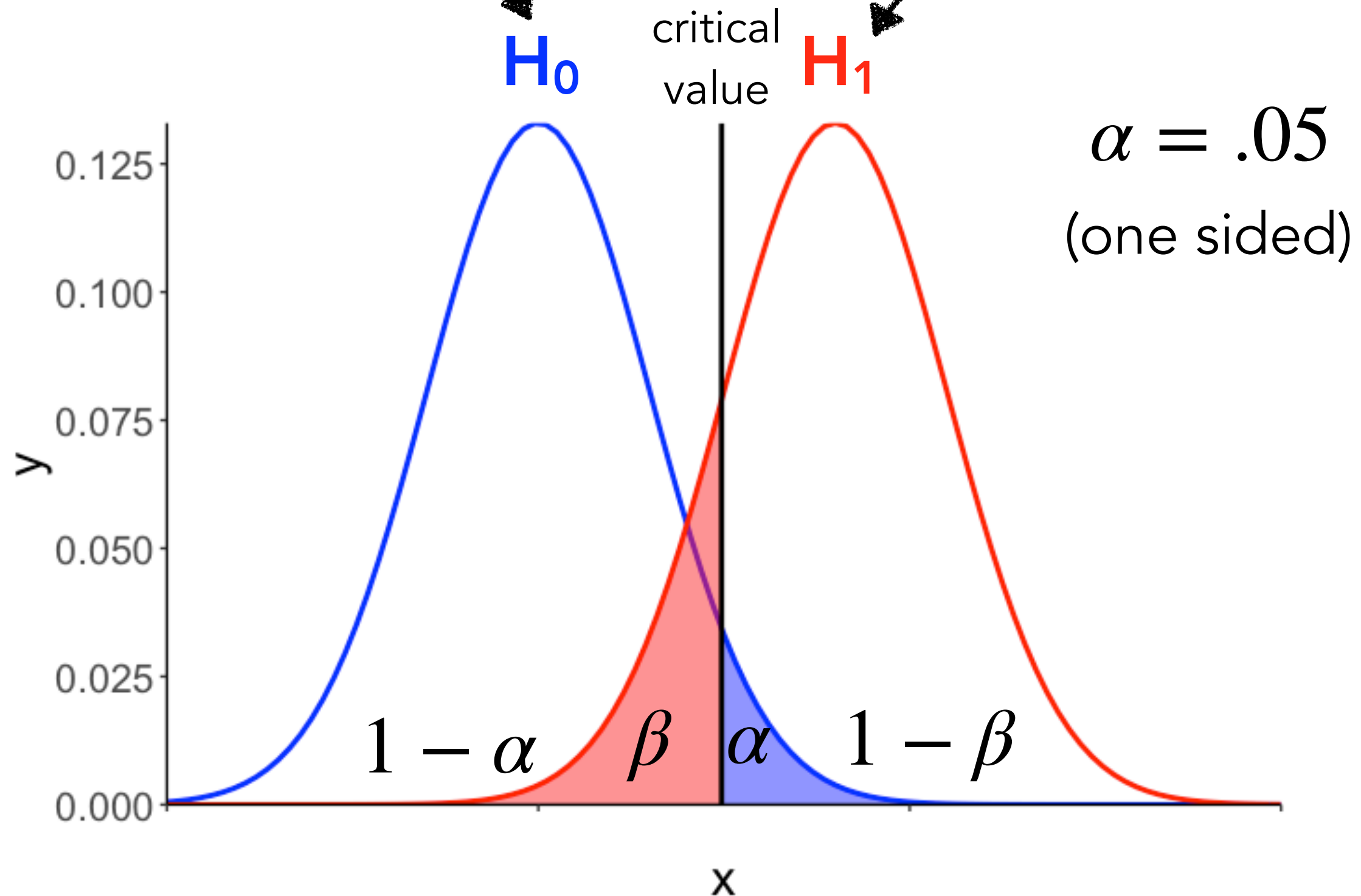


What affects power?

false positive	Type I error $p(\text{reject } H_0 H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 H_1 \text{ is true})$	Power $1 - \beta$
false negative	Type II error $p(\text{not reject } H_0 H_1 \text{ is true})$	β	true negative	Specificity $p(\text{not reject } H_0 H_0 \text{ is true})$	$1 - \alpha$

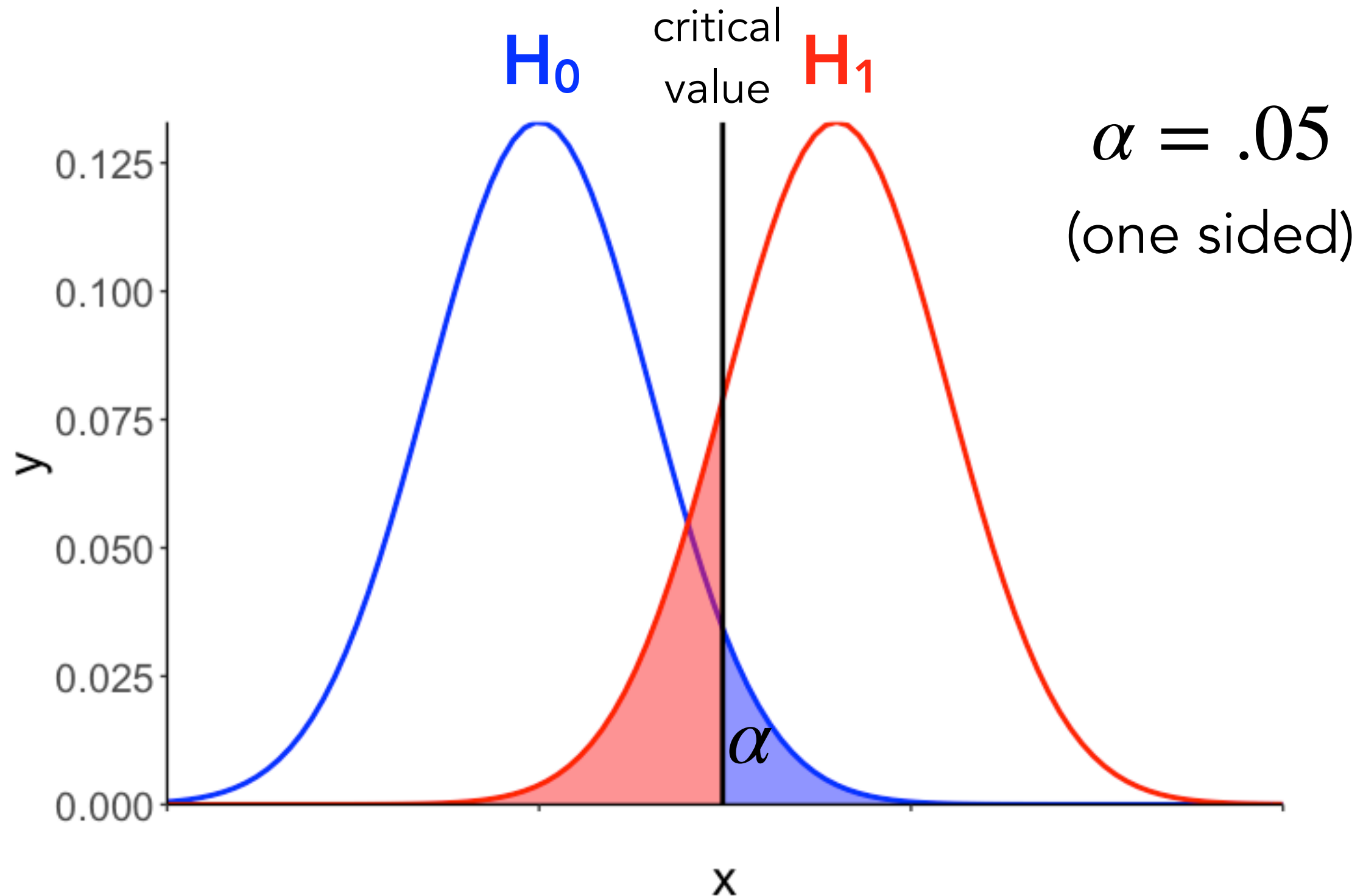
sampling distribution if H_0 is true

sampling distribution if H_1 is true



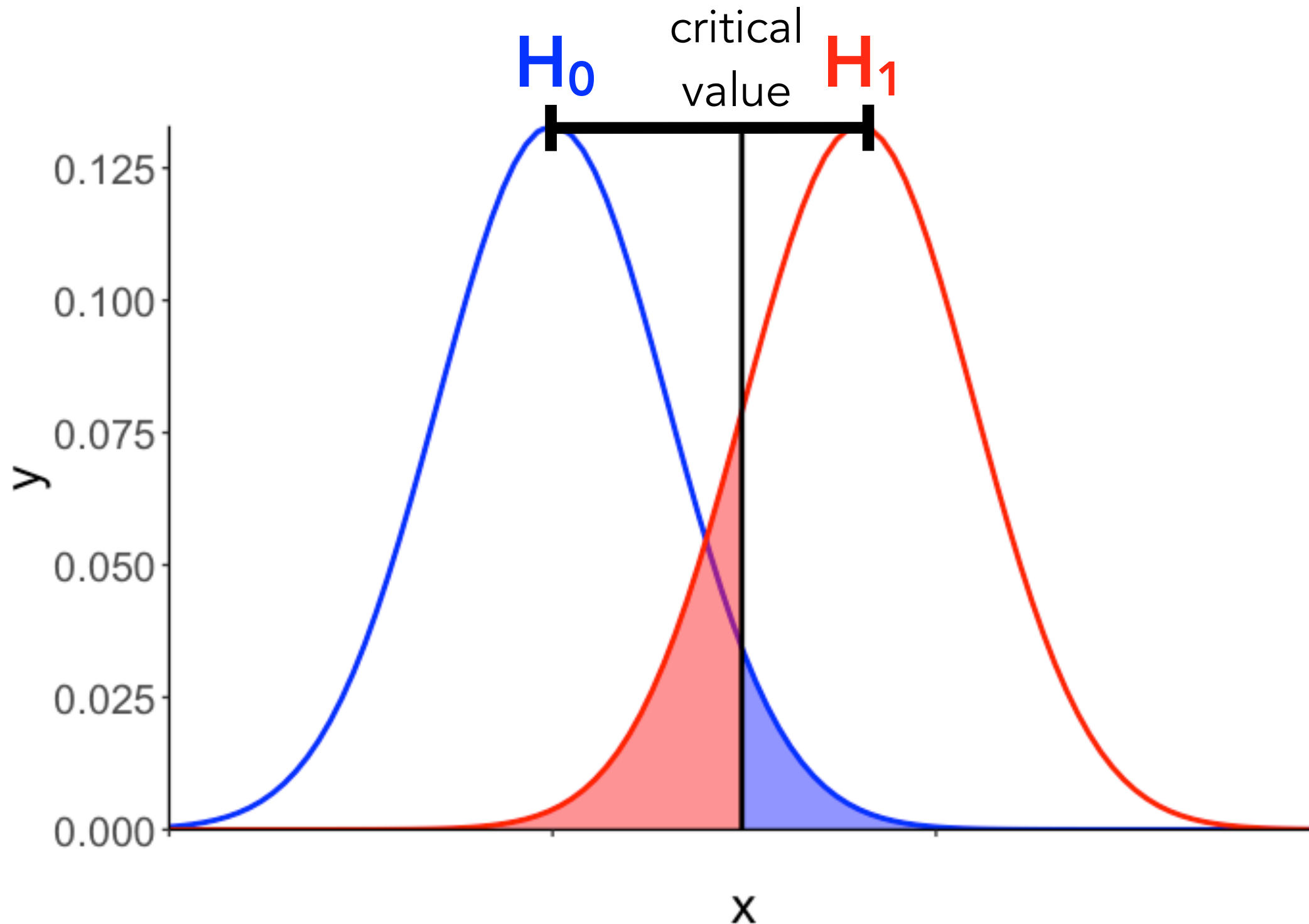
α affects power
(e.g. p-value threshold)

false positive	Type I error $p(\text{reject } H_0 H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 H_1 \text{ is true})$	Power $1 - \beta$
false negative	Type II error $p(\text{not reject } H_0 H_1 \text{ is true})$	β	true negative	Specificity $p(\text{not reject } H_0 H_0 \text{ is true})$	$1 - \alpha$



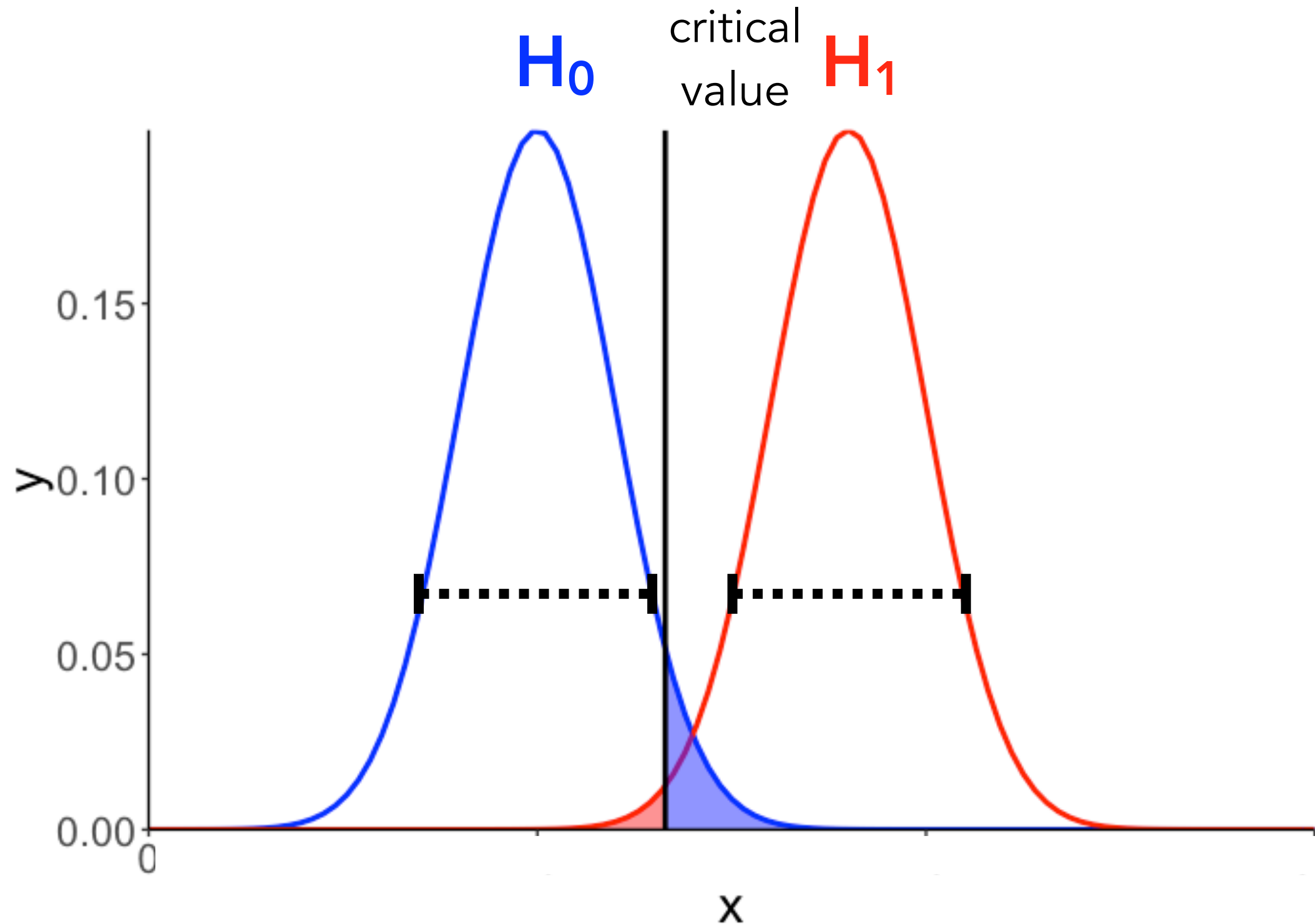
distance between
means affects power

false positive	Type I error $p(\text{reject } H_0 H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 H_1 \text{ is true})$	Power $1 - \beta$
false negative	Type II error $p(\text{not reject } H_0 H_1 \text{ is true})$	β	true negative	Specificity $p(\text{not reject } H_0 H_0 \text{ is true})$	$1 - \alpha$



variance affects power

false positive	Type I error $p(\text{reject } H_0 H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 H_1 \text{ is true})$	Power $1 - \beta$
false negative	Type II error $p(\text{not reject } H_0 H_1 \text{ is true})$	β	true negative	Specificity $p(\text{not reject } H_0 H_0 \text{ is true})$	$1 - \alpha$



Effect sizes

Effect sizes

- a p-value tells us whether we can reject the H_0
- effect sizes is a measure of the strength of the actual effect

Why can't we just use p-values as a measure of the effect size?

$$F = \frac{\text{PRE}/(\text{PA} - \text{PC})}{(1 - \text{PRE})/(n - \text{PA})}$$

PRE = proportional reduction in error

PA = # parameters in the augmented model

PC = # parameters in the compact model

n = sample size

any PRE will become significant if n gets large enough

Effect sizes

- a p-value tells us whether we can reject the H_0
- effect sizes is a measure of the strength of the actual effect

Why can't we just use p-values as a measure of the effect size?

$$t = \frac{\hat{\beta}}{SE_{\hat{\beta}}} = \frac{\hat{\beta}}{\frac{SD_{\hat{\beta}}}{\sqrt{n}}}$$

any t will become significant if n gets large enough



Remember!

- Large t-stat or small p-value is **not** about what variables are most important for prediction!
- It's about what **signals** can be discerned from **noise**
 - *Does the addition of this parameter provide lower prediction error than what adding random noise would produce?*
- Why? Because our *uncertainty* (SE) is influenced by
 - Magnitude of $(\hat{\beta})$
 - How bad model is (MSE)
 - Effective sample size (# observations vs # params aka DF)
 - Variance of each predictor ($x_1 \dots x_n$)
 - Correlation between predictors (columns of X)

Statistical significance is not practical significance

Effect Size Reference

Guide to Effect Sizes and Confidence Intervals

Table of Contents



Home

- 1 Defining Effect Sizes
- 2 Benchmarks
- 3 Reporting Effect Sizes
- 4 Interpreting Confidence Intervals
- 5 Reporting Confidence Intervals
- 6 Using R
- Standardized Effect Sizes** ▾
 - 7 Mean Differences
 - 8 Correlations
 - 9 Categorical/Proportional Data
 - 10 ANOVAs
 - 11 Variability Differences
 - 12 Non-Parametric Effect

Guide to Effect Sizes and Confidence Intervals

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Table of contents

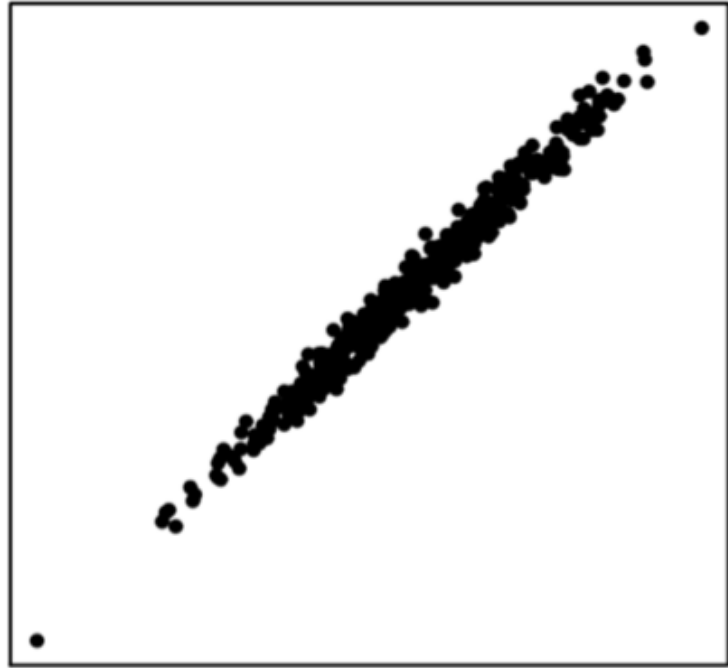
[Welcome](#)

[Introduction](#)

<https://matthewbjane.quarto.pub/>

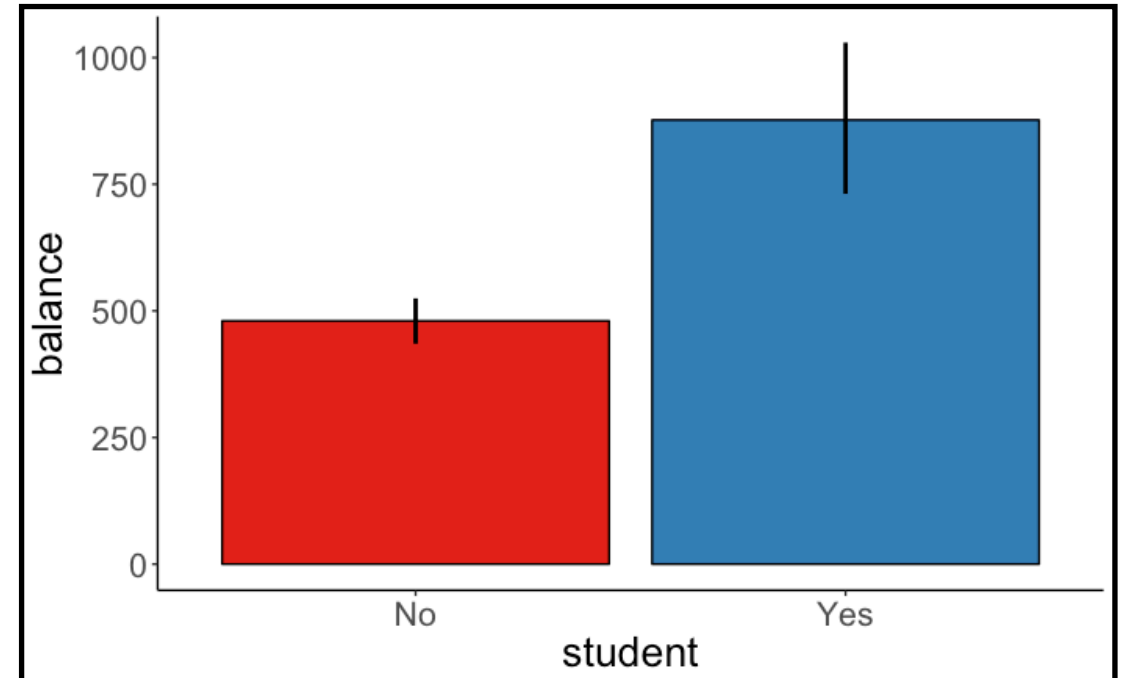
Common effect sizes

Relationships between variables



r correlation

Differences between groups



Cohen's d

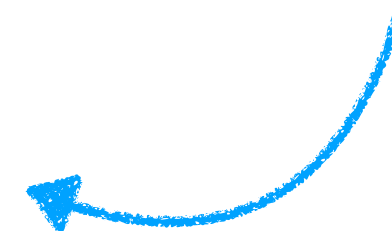
Pearson correlation: we've seen this before!

$$r(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \cdot \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

Cohen's guidelines for the social sciences

Effect size	r
Small	0.1
Medium	0.3
Large	0.5

depends very
much on the
domain



Cohen's d

- standardized difference between two means

absolute
difference
between means

$$d = \frac{|\bar{y}_1 - \bar{y}_2|}{s_p}$$

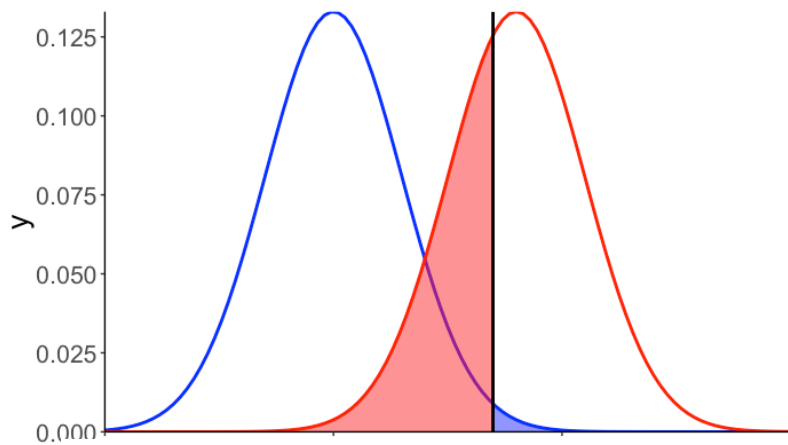
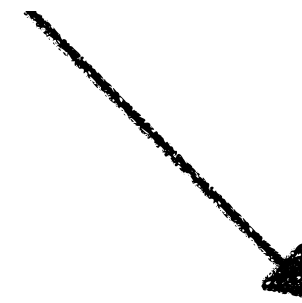
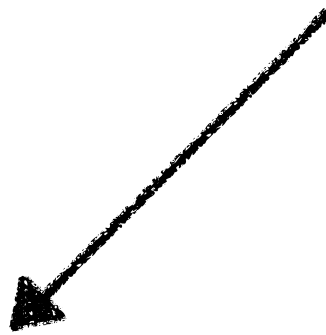
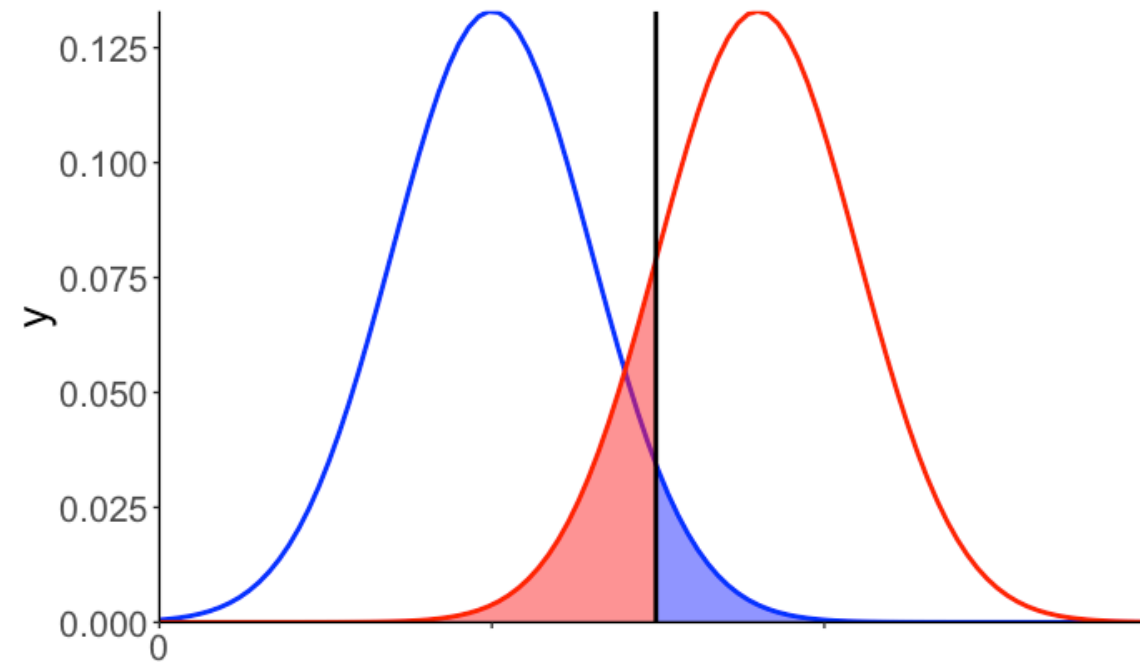
pooled standard variation

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

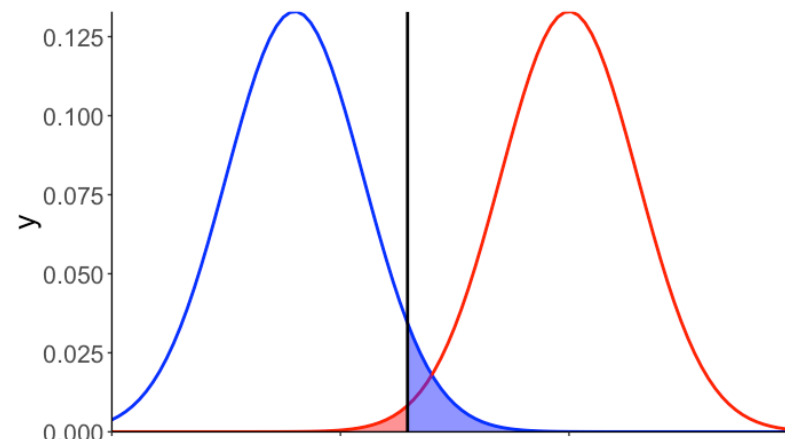
<i>Effect size</i>	<i>d</i>
Very small	0.01
Small	0.20
Medium	0.50
Large	0.80
Very large	1.20
Huge	2.0

Ratio of: between variation / pooled variation

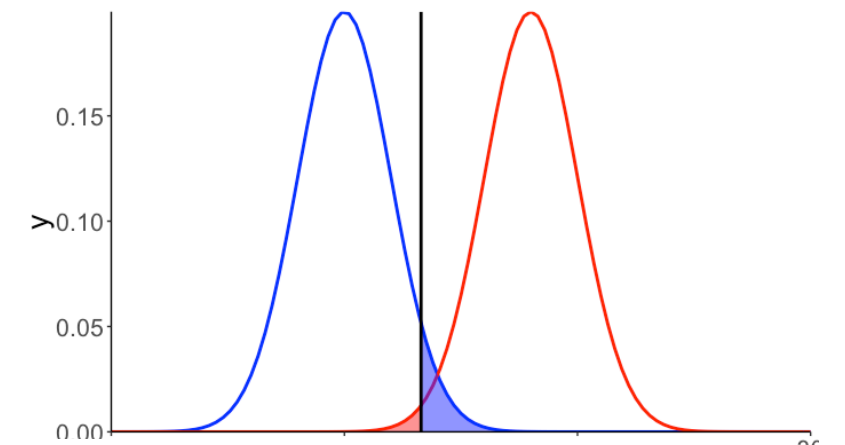
The knobs we can turn to affect power



a threshold



effect size



sample size

How do we “turn the knobs” (determine power)?

Option 3

Power analysis via **simulation**

- “4 horsemen” of computational statistics (wk2 01-15)
- **Monte-carlo simulation**

- Bootstrapping
- Permuting
- Cross-validation

Recipe

- pick: α , n , *effect-size*
- simulate a large number of data sets of size n with the specified *effect-size*
- for each data set, run a statistical test to calculate the *p-value*
- calculate **sensitivity/power**: proportion of times we *correctly rejected* H_0 (given that H_1 is true)

On your own...

- Grab github classroom link from course website
- Work through
 - 01_ReadingQs
 - 02_Power
- You will have additional time tomorrow in class
- **Deadline:** Monday